

Uncertainty area	Why it matters	How it affects the comparison	How it will be handled
Service life	The real service life of frames, glazing, seals and hardware depends on material quality, exposure, maintenance and installation quality	If service life is shorter than assumed, repair benefits may be reduced; if longer, full replacement may be less justified	Use low, baseline and high assumptions instead of one fixed value
Repair frequency	Scenario B depends on how often smaller components need repair or replacement	Too many repair visits may increase transport, labour and cost; too few may overstate the feasibility of product-life extension	Use 1, 2 and 3 repair-event assumptions
Spare-part availability	Component replacement only works if compatible parts remain available	If compatible parts are unavailable, a technically repairable window may still be replaced fully	Treat as an implementation-gap risk
Component access	Repair may be technically possible but practically difficult if hinges, seals, glazing or hardware cannot be reached	Poor access can turn Scenario B from repair-based life extension into premature full replacement	Treat access as a key enabling condition
Transport	Replacement and repair both require transport, but the number and distance of trips differ	Long-distance repair, specialist service or end-of-life transport may reduce expected benefits	Use low, baseline and high transport-distance assumptions
Operational performance	Replacement may improve thermal performance, while repair may retain existing performance unless glazing is upgraded	Scenario A may look better if operational savings are large; Scenario B may look better if embodied impacts dominate	Keep comparison qualitative unless reliable operational data are available
Embodied impacts	Full replacement adds a complete new window system, while repair adds smaller components and labour	Scenario A may create higher production-stage burden; Scenario B may still create burden through repeated components and service visits	Discuss as qualitative burden categories, not exact values
End-of-life treatment	Recycling, landfill, downcycling or uncertain recovery can change the apparent benefit of replacement or repair	Environmental burden may shift to end-of-life stages if recovery routes are unclear	Treat as burden-shifting uncertainty
Reuse feasibility	Removed windows are often made to measure and difficult to reuse directly	Scenario C is not robust enough for the main comparison	Exclude direct reuse from main scenario comparison and mention as future research
Data availability	Product-specific use-phase, repair-event and end-of-life data are incomplete	Exact environmental savings cannot be claimed	State clearly that this is a lifecycle scenario comparison, not a complete LCA